Lab 9

IR Absorption by Greenhouse Gases

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Equipment List

- Ruler
- Digital Thermometer
- 250 mL beaker
- 500 mL beaker
- Hot Plate
- Lab jack
- Heatlamp with attachment to table
- IR thermal imaging camera
- IR thermometer
- plastic basket with windows
- ziploc and "food storage" bags (for classroom)

9.1 Introduction and Definitions

9.1.1 Climate Sensitivity

This week in class we are attempting to identify the precise connection between additional greenhouse gases in the atmosphere and the resulting temperature increase. This connection is called the **climate sensitivity**[1]. In lecture and discussion, we will attempt to see this connection through the deepening and broadening of CO_2 absorption lines in the blackbody spectrum of the Earth surface's infrared radiation. Here, in lab, we will simply try to quantitatively observe the absorption of infrared light by CO_2 (and possilbly air and water vapor).

9.1.2 Absorption lines and IR thermometer/camera measurements

Previously in Discussion we learned how to draw blackbody spectra on log-log paper and then, when studying the IR absorption by greenhouse gases, we sketched absorption lines on these

[1] *Climate sensitivity* is often used as a technical term by climate scientists to refer to "the temperature increase due to a doubling of atmospheric CO_2"; but it is also used in this more general sense.

spectra. Recall the relevant equations for this process:

$$\lambda_{\text{peak}} = \frac{2900\ \mu\text{m}\cdot\text{K}}{T} \quad ; \quad F = \left(5.7\times10^{-8}\,\frac{\text{J/s}}{\text{m}^2\cdot\text{K}^4}\right)T^4 \quad ; \quad F_\lambda = \frac{F}{1.5\times\lambda_{\text{peak}}}$$

To create the blackbody curve, we drew the peak ("hat") with three points at wavelengths of $[\frac{\lambda_p}{2}, \lambda_p, 3\lambda_p]$, and at spectral flux values of $[\frac{F_\lambda}{10}, F_\lambda, \frac{F_\lambda}{10}]$; the curve dropped off with a slope of 4 to high wavelengths, and more quickly at low wavelengths.

On a single piece of log-log graph paper, draw blackbody curves for the following temperatures: room temperature (20°C = 293 K); hot water (50°C = 323 K); boiling water (100°C = 373 K); and a heat lamp at 450°C = 673 K. Make your calculations here:

We will attempt to see the absorption of infrared light by carbon dioxide using IR thermometer guns and imaging cameras, but this is not a completely straightforward process because of how these devices work. First, they estimate temperature by collecting all light over a range of wavelengths, 8–14 μm. They measure the total flux emitted by an object in that wavelength range and then compare it to what would be expected from blackbody curves at different temperatures (the "area under" the curve between these wavelengths). On your blackbody graph, draw two vertical lines to indicate this range of wavelengths. Looking at your curves, which of the objects would have the most flux in this range. Can you estimate how much more[2] it is than the second most?

[2] An approximation to the observed total flux is the area of a rectangle that has height F_λ (at the center of the 8–14 μm interval) and width: 14 μm − 8 μm = 6 μm.

A second issue with these devices is that their wavelength range has been specifically chosen to *avoid* the CO_2 absorption lines, which you remember from our work in Discussion are at 4.3 μm and 15 μm. Draw vertical lines on your blackbody curve to indicate these wavelengths. Is it possible that a carbon dioxide absorption line will affect anything in the 8–14 μm sensitivity range of the devices? Why or why not?

We do expect to see some small effect because of the line width of carbon dioxide, particularly at 15 μm. As that line takes a bite out of the spectrum, it should cover a small part of the sensitivity range of the devices. Sketch an absorption "valley" on each of your spectra at 15 μm. Looking at your graphs, do you expect that any of the three non-room-temperature sources would be more affected than others?

9.2 Experiments and Analysis

Safety Notes:

- We have tanks of CO_2 and N_2 gas in the classroom. Do not disturb their buckle attachments to the table as the greatest danger would be for them to fall, break a valve and then "rocket" across the room. Be gentle when screwing the valves to release the gases.

As always: read the full procedures prior to beginning any experimentation.

9.2.1 IR transparency of plastic bags

Make some measurements and observations to test transparency of the plastic bags we will later fill with gas. There are multiple types of bags that can be used (multiple "Ziploc"-style, and food-storage "produce" bags). You can use IR thermometer gun and the IR imaging camera. Observe what is visible in the camera image and how the measurements of temperature recorded by both devices changes through one or multiple plastic bag layers. You can use humans as sources of IR or fill a beaker with "hot" tap water.

Record your observations, measurements, and conclusions here.

9.2.2 IR absorption in air

Complete these steps in a group of four using your table's Mileseey infrared camera. We are interested in the device's measured temperature of the vat of water at different distances, but because it is cooling in the room over time, we need to observe that cooling for a few minutes at each distance and then extrapolate backwards to obtain an estimate of the initial temperature for each distance.

Procedure:

- Starting at the vat of hot water (tap water at $\sim 50°C$), use a meter stick (located along the walls on top of the power strip) to mark out distances of 1 m, 2 m, 3 m, 4 m, and 5 m with masking tape on the floor.

- Start your stopwatch.
- Use the IR camera to make measurements of the temperature of the vat of water from zero to five meters, recording the time, distance, and temperature. You should be able to use the (red) "max" temperature value shown on the image (or the white central cross-hairs temperature shown at top left). These measurements will probably be 20–30 seconds apart. Enter these values in a three-column table in Google Sheets, recording the time in two columns, minutes and seconds, so that you can later convert to "t (s)".
- Repeat the zero to five meter measurements 5–7 times.

Analysis: Copy the Google Sheets table for each student. Using the two time columns, create a "time in seconds" column (with an excel command like "`=A2*60+B2`"). On a single sheet of graph paper, plot six data sets for the Temperature vs. time values at each distance (make each axis range as narrow as possible while still showing the data . . . it is not necessary to show the "origin"). [Alternatively you could copy your data into separate sheets for each distance and insert Charts for each.] Using a ruler, draw a line of best fit and obtain the "y intercept," i.e., the temperature at time zero, for each of the datasets. Then make a new plot, on another sheet of graph paper, showing initial temperature vs. distance.

Briefly state what you observe in each of the two graphs. How do your results demonstrate absorption of IR light, if at all? If there is IR absorption, what do you think is the primary contributor?